

Dynamical Low-rank.

rank.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} \rightarrow \text{rank-one matrix.}$$

$$\text{decided by } \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} (1 \ 2 \ 3) = u v^T$$

(Matrix is 2D but the 2 dimensions are highly relevant)

rank = r ?

$$A = u_1 v_1^T + u_2 v_2^T + \dots + u_r v_r^T = \underset{m \times r}{U} \underset{r \times r}{S} \underset{r \times n}{V^T}$$

$$\text{SVD } A = U S V^T \quad (\sigma_1 u_1 v_1^T + \dots + \sigma_p u_p v_p^T)$$

$$\sigma_1 \geq \dots \geq \sigma_p \geq 0$$

$$\text{if } \sigma_1=100, \sigma_2=30, \sigma_3=5, \sigma_4=0.01, \sigma_5=0.001$$

$$\Rightarrow \text{cut } A \approx \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \sigma_3 u_3 v_3^T$$

$$\text{PDE: } A \rightarrow f(x, v)$$

$$\text{Discretize } x, v \Rightarrow \begin{cases} x_1, \dots, x_{N_x} \\ v_1, \dots, v_{N_v} \end{cases}$$

$$\Rightarrow F_{ij} = f(x_i, v_j)$$

$$\text{If } f(x, v) = p(x) M(v)$$

$$\text{Then } F = \begin{pmatrix} p(x_1) \\ \vdots \\ p(x_{N_x}) \end{pmatrix} (M(v_1) \dots M(v_{N_v})) \quad (\text{rank one})$$

Paper: A review of low-rank

Abstract

$$\text{Time-dependent kinetic models. } (\mathbb{R}^3 \times \mathbb{R}^3 \xrightarrow[\mathbb{R}^3]{f(t, x, v)})$$

1. DLR Dynamical Low-rank.

2. SAT Step-and-Truncate.

$$f(t, x, v) \stackrel{?}{=} \sum_{i=1}^N \sum_{j=1}^N U_i(t, x) S_{ij}(t) V_j(t, v)$$

Sec 1 Relevance & challenges of kinetic simulation

$$\text{Def } f(t, x, v)$$

$$\text{Massive density} = \rho(t, x) = m_p \int f(t, x, v) dv$$

$$\text{Density} = n(t, x) = \int f(t, x, v) dv$$

$$\text{Kinetic equation: } \underbrace{\partial_t f(t, x, v)}_{\text{Transport}} + \underbrace{v \cdot \nabla_x f(t, x, v)}_{\text{Influence from outside}} + \mathcal{J}(f, \nabla_v f) = C(f) \quad (*)$$

LHS. streaming operator

RHS. Collision operator

Hard to solve (*) why?

1. High-D $100^6 = 10^{12}$

2. Collision operator complex

$\left\{ \begin{array}{l} \text{Design fast-numerical algorithm} \\ \text{Simplified model} \\ \dots \end{array} \right.$

High-D : $f(t, x, v) \approx \sum_{i=1}^r \sum_{j=1}^r U_i(t, x) S_{ij}(t) V_j(t, v)$
 — Not mesh-free. (different for Monte-Carlo)

Sec 2 Low-rank approximation

- Requirement
1. Find a method to express the solution, degrees of freedom ↓
 2. High-efficiency — updated as time

Low-rank representation

$$f(t, x, v) = \sum_{i=1}^r \sum_{j=1}^r U_i(t, x) S_{ij}(t) V_j(t, v)$$

i.e. $f = USV^T$

$$O(n^6) \rightarrow O(rn^3) \quad (r \ll n^3)$$

Compared with Schmidt

Schmidt: $f(x, v) = \sum_{i=1}^r \sigma_i U_i(x) V_i(v) \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$

SVD: $S \in \mathbb{R}^{r \times r}$

⇒ why original problem low-rank?

1° Collision ↑ → { fluid limit
diffusive limit.

$$\partial_t f + \text{trans} = \frac{1}{\varepsilon} C(f)$$

↳ Collision ↑ ⇒ $\varepsilon \rightarrow 0 \Rightarrow C(f) = 0$

Diffusion limit

Collision → Isotropic

$$f(t, x, v) = p(t, x) + \varepsilon g(t, x, v)$$

$\frac{1}{\varepsilon}$ 碰撞强度

$$\varepsilon \rightarrow 0 \quad f \rightarrow p$$

$$f(t, x, v) = U_i(t, x) V_i(x) \quad U_i = p, \quad V_i = 1$$

$$f \approx p(x) + \varepsilon \sum_{i=1}^3 a_i(x) b_i(x) + O(\varepsilon^2) \quad (\text{rank} \leq 4)$$

Fluid limit

$$\text{Maxwellian: } f(t, x, v) = \exp\left(-\frac{(v - u(t, x))^2}{2\theta(t, x)}\right) (p(t, x) + \varepsilon g(t, x, v))$$

$$\varepsilon \rightarrow 0 \quad \varepsilon g(t, x, v) \rightarrow 0.$$

$$M(x, v) = p(x) \exp(\quad) = (\text{Taylor})$$

2° Collisionless

e.g. $\partial_t f(t, x, v) + v \cdot \nabla_x f(t, x, v) = 0. \quad (\text{free trans})$

$$f(t, x, v) = \exp(ik \cdot x) \exp\left(-\frac{v^2}{2}\right) \quad (\text{rank } 1)$$

$$\text{Characteristic} \Rightarrow f(t, x, v) = \exp(ik \cdot (x - vt)) \exp\left(-\frac{v^2}{2}\right) \quad (\text{rank } 1)$$

free-trans ⇒ weak nonlinear (Vlasov-Maxwell Equation)

$$\text{Prop: } f(t=0) \text{ rank } r \Rightarrow \text{Linear solution rank} \leq r+1$$

Sec 3 Efficient time evolution of low-rank solutions

$$f(t, x, v) \approx U(t, x) S(t) V(t, v)^T \quad t_n \xrightarrow{?} t_{n+1}$$

Dynamical Low Rank, DLR. (restrict on Low-rank)

Step-and-Truncate, SAT. rank $r \downarrow$

$$\mathcal{M}_r = \{f = USV^T : U^T U = 1, V^T V = 1, S \text{ invertible}\}.$$

Example: $\partial_t f = v \partial_x f + E \partial_v f \stackrel{\text{def}}{=} \text{RHS}(f)$
 $= -v \partial_x f + E \partial_v f$

$$E \begin{cases} -\partial_{xx} \phi = P = 1 - f dv \\ E(x, t) = -\partial_x \phi \end{cases}$$

DLR: fix rank = r

$$f(t, x, v) = \sum_{i=1}^r \sum_{j=1}^r U_i(t, x) S_{ij}(t) V_j(t, v)$$

$$\langle U_i, U_k \rangle_x = \delta_{ik} \quad \langle V_j, V_l \rangle_v = \delta_{jl}$$

$$\mathcal{M}_r = \{f : \text{rank}(f) = r\}$$

Trans direction: $\text{RHS}(f) \rightarrow P(f) \text{RHS}(f)$

↓
投影

$$P(f)g = P_U g + P_V g - P_U P_V g$$

$$\partial_t f = -v \partial_x f + E \partial_v f$$

$$\Rightarrow \frac{\partial f}{\partial t} + P(f) \left(v \frac{\partial f}{\partial x} - E \frac{\partial f}{\partial v} \right) = 0$$

$$\text{RHS}_{LR} = \text{RHS} \left(\sum_{k,l} U_k S_{kl} V_l \right)$$

$$f = USV^T \Rightarrow \partial_t f = (\partial_t U) S V^T + U (\partial_t S) V^T + US (\partial_t V^T)$$

$$\xrightarrow{?} (\text{RHS})_{LR}$$

Gauge condition $\langle U_i, \partial_t U_k \rangle_x = 0 \quad \langle V_j, \partial_t V_l \rangle_v = 0$

$$f(t, x, v) = \sum_i \sum_j U_i(t, x) S_{ij}(t) V_j(t, v)$$

$$\partial_t f = \sum_{ij} (\partial_t U_i) S_{ij} V_j + \sum_i U_i (\partial_t S_{ij}) V_j + \sum_i U_i S_{ij} (\partial_t V_j) \quad (C)$$

Project (C) to U_i and V_j

$$\Rightarrow \langle U_i V_j, \partial_t f \rangle_{xv} = \partial_t S_{ij}$$

$$\partial_t S_{ij} = \langle U_i V_j, \text{RHS}_{LR} \rangle_{xv} \quad (6)$$

$$U_i: \sum_j S_{ij} \partial_t V_j = \langle U_i, \text{RHS}_{LR} \rangle_x - \sum_j (\partial_t S_{ij}) V_j \quad (7)$$

$$V_j: \sum_i S_{ij} \partial_t U_i = \langle V_j, \text{RHS}_{LR} \rangle_v - \sum_i U_i (\partial_t S_{ij}) \quad (8)$$

(6) updates S

(7) updates V_j

(8) updates U_i

Effective computing

$$\langle U_i V_j, \text{RHS}_{LR} \rangle_{xv} = - \sum_{k,l} \langle U_i \partial_x U_k \rangle_x S_{kl} \langle V_j V V_l \rangle_v + \sum_{k,l} \langle U_i E U_k \rangle_x S_{kl} \langle V_j \partial_v V_l \rangle_v$$

$$\langle U_i, \text{RHS}_{LR} \rangle_x = - \sum_{k,l} \langle U_i \partial_x U_k \rangle_x S_{kl} V_l + \sum_{k,l} \langle U_i E U_k \rangle_x S_{kl} \partial_v V_l$$

$$\langle V_j, \text{RHS}_{LR} \rangle_v = - \sum_{k,l} \partial_x U_k S_{kl} \langle V_j V V_l \rangle_v + \sum_{k,l} E U_k S_{kl} \langle V_j \partial_v V_l \rangle_v$$

Difficulties in solving (6) - (8)

S^{-1} ?

1° Projector Splitting Integrator

$$P = P_U + P_V - P_U P_V$$

$$\partial_t f = P_U \text{RHS}(f)$$

$$\partial_t f = P_V \text{RHS}(f)$$

$$\partial_t f = -P_U P_V \text{RHS}(f)$$

K-step $\Rightarrow$ S-step $\Rightarrow$ L-step

$$K_j(x) = \sum_i U_i(x) S_{ij} \quad L_i(v) = \sum_j S_{ij} V_j(v).$$

$$\text{i.e. } K = US \quad L = VS^T$$

$$K-S-L: \quad f^n = U^n S^n (V^n)^T \quad t_n \rightarrow t_{n+1}.$$

1. K-step: fix V , update X .

$$\partial_t K_i(t, x) = \langle \text{RHS}_K(K(t, x), V^n), V_i^n \rangle_V$$

$$K_i(t_n, x) = \sum_j U_j^n(x) S_{ji}^n$$

Integrate to t_{n+1} . Then QR: $K(t_{n+1}) = U^{n+1} S^*$

2. S-step: $\partial_t S_{ij}(t) = -\langle \text{RHS}_S(U^{n+1}, S(t), V^n), U_i^{n+1} V_j^n \rangle_{XV}$

$$S(t_n) = S^*$$

$$\Rightarrow S^{**} = S(t_{n+1})$$

3. $\partial_t L_i(t, v) = \langle \text{RHS}_L(U^{n+1}, L(t, v)), U_i^{n+1} \rangle_X$

$$L_i(t_n, v) = \sum_j S_{ij}^{**} V_j^n(v)$$

$$\Rightarrow L_i(t_{n+1}, v) \stackrel{\text{QR}}{=} V^{n+1} S^{n+1 T}$$

2° Basis update & Galerkin (BUG) Integral

Fix rank - BUG

K-step $\rightarrow U^{n+1}$

L-step $\rightarrow V^{n+1}$

$\Rightarrow$ Galerkin S-step.

Augmented BUG

$$\hat{U} = \text{orth}(L K^{n+1}, U^n), \quad \hat{V} = \text{orth}(L L^{n+1}, V^n)$$

K-step: $\partial_t K_i = \langle \text{RHS}_K(K, V^n), V_i^n \rangle_V$

$$K_i(t_n) = \sum_j U_j^n S_{ji}^n$$

L-step: $\partial_t L_i = \langle \text{RHS}_L(U^n, L), U_i^n \rangle_X$

$$L_i(t_n) = \sum_j S_{ij}^n V_j^n$$

Augmented $\hat{U}, - = \text{QR}(L K^{n+1}, U^n)$

$$\hat{V}, - = \text{QR}(L L^{n+1}, V^n)$$

Update S: $\partial_t \hat{S}_{ij} = \langle \text{RHS}_S(\hat{U}, \hat{S}, \hat{V}), \hat{U}_i \hat{V}_j \rangle$

$$\hat{S}(t_n) = M S^n N^T$$

$$M_{ij} = \langle \hat{U}_i, U_j^n \rangle_X \quad N_{ij} = \langle \hat{V}_i, V_j^n \rangle_V$$

SVD: $\hat{S}(t_{n+1}) = P \Sigma Q^T, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_{2r})$

$$\text{Choose min } r_{n+1} \text{ s.t. } \sum_{j=r_{n+1}+1}^{2r_n} \sigma_j^2 \leq \epsilon^2$$

$$S^{n+1} = \text{diag}(\sigma_1, \dots, \sigma_{r_{n+1}}), \quad U^{n+1} = \hat{U} P_{:, 1:r_{n+1}} \quad V^{n+1} = \hat{V} Q_{:, 1:r_{n+1}}$$

3.2 SAT.

why rank ↑.

$$f^n(x, v) = \sum_{j=1}^{r_n} U_j^n(x) S_j^n V_j^n(v)$$

Upwind Euler: $f^{n+1} = f^n - \Delta t (V \partial_x f^n + E^n(x) \partial_v f^n)$

$$\Rightarrow f^{n+1} = \sum_{j=1}^{r_n} S_j^n [U_j^n V_j^n - \Delta t (\partial_x U_j^n)(V V_j^n) - \Delta t (E^n U_j^n)(\partial_v V_j^n)]$$

$$\text{rank} \leq 3r_n$$

$$\tilde{f}^{n+1} = \tilde{U} \tilde{S} \tilde{V}^T$$

QR for UV, SVD. $\Rightarrow r_{n+1}$

Sec 4. Phase space and temporal discretizations

$$\partial_t f = \text{RHS}(f)$$

$$\partial_x, \partial_v ?$$

$$F_{kl}^n = f(t_n, x_k, v_l)$$

$$F^n = U^n S^n (V^n)^T$$

1° Phase-space discretization

$$\partial_t f = -V \partial_x f + E(x) \partial_v f =: \text{RHS}(f)$$

$$\text{RHS}(U_j V_j) = -(\partial_x U_j)(V V_j) + (E U_j)(\partial_v V_j)$$

$$F^{n+1, \star} = F^n + \Delta t \text{RHS}(F^n)$$

Upwind principle: $\partial_t f + a \partial_x f = 0.$

Information $\rightarrow a$

$a > 0$ (from left)

$$\Rightarrow \partial_x U_j(V V_j) \Rightarrow (D_x^+ U_j) \otimes (V^+ * V_j) + (D_x^- U_j) \otimes (V^- * V_j)$$

...

Flux-difference form

$$(\partial_x q)_k \approx \frac{\hat{q}_{k+1/2} - \hat{q}_{k-1/2}}{\Delta x}$$

Second-order $\rightarrow$ central difference

On boundary $\partial \Omega_x \times \Omega_v$

$v \cdot n(x) < 0$ in $\Rightarrow$ set boundary conditions

$v \cdot n(x) > 0$ out. ~~X?~~

How to deal with inflow?

SAT: $S_{\text{exp}} + f_{\text{in}} \rightarrow \text{Truncate.}$

DLR: $f_{\text{in}} \notin \text{Span}\{V_1, \dots, V_r\} \rightarrow \dots$

2° Temporal discretization

DLR: $\partial_t K = R_K(K) \quad \partial_t L = R_L(L) \quad \dot{S} = R_S(S).$

$K^{n+1} = K^n + \Delta t R_K(K^n)$ or $R_K(K^{n+1})$ (Implicit, better due to lower rank)

SAT: $F^{n+1} = F^n + \Delta t R_h(F^n)$ ($R_h(F^{n+1})$ difficult to calculate)

IMEX: $F^{n+1} = F^n + \Delta t R_{\text{nonstiff}}(F^n) + \Delta t R_{\text{stiff}}(F^{n+1})$

3° Semi-Lagrangian (Later)

4° Stability

Sec 6 Structure keep

1° Conservation laws

2° Asymptotic preservation.

3° Positivity

1° Conservation Laws.

Flux:

$$\text{Partial: } p(t, x) = \int f dv$$

$$J(t, x) = \int v f dv.$$

$$K(t, x) = \frac{1}{2} \int |v|^2 f dv$$

$$e(t, x) = K(t, x) + \frac{1}{2} |E(x)|^2$$

Why normal non-rank x

$1(v) \notin \text{span}\{v_j\}$

$$\text{Conservation Law: } p = \int f dv \quad J = \int v f dv \quad K = \frac{1}{2} \int |v|^2 f dv.$$

$$\partial_t p + \partial_x J = 0$$

$$\partial_t J + \partial_x \sigma = 0$$

$$\partial_t e + \partial_x Q = 0.$$

$$\sigma = \int v \otimes v f dv \quad Q = \frac{1}{2} \int |v|^2 v f dv$$

Fix for SAT: (mainly caused by SAT)

1. conventional truncation

keep $\text{span}\{1, v, v^2\}$. using SVD only for the others.

2. LoMac (Local Macroscopic Conservative, LoMac)

$$\text{SAT: } f^n \rightarrow f_*^{n+1} \xrightarrow[\text{Conservation?}]{\text{Truncate}} f^{n+1}$$

$$\text{e.g. } \int f_* dv = \int f dv.$$

$$\text{LoMac } \begin{cases} f(x, v) \\ p(x), J(x), e(x). \end{cases}$$

$$f \rightarrow \sigma, Q, \text{ boundary flux} \rightarrow p, J, e.$$

KFVS: $f \rightarrow$ Form interface numerical fluxes for macroscopic conservation equations

$$\Rightarrow \begin{cases} f_*^{n+1} & (\text{Lo}) \\ p_{\text{mac}}^{n+1}(x), J_{\text{mac}}^{n+1}(x), e_{\text{mac}}^{n+1}(x) & (\text{Mac}) \end{cases}$$

$$f = \underbrace{f_{\text{macro}}}_{p, J, K} + f_{\text{micro}}$$

$$\Rightarrow f_*^{n+1} = p_{\text{macro}} f_*^{n+1} + g_*^{n+1} \rightarrow (1-p) f_*^{n+1}$$

$$\int g_* dv = 0 \quad \int v g_* dv = 0 \quad \int v^2 g_* dv = 0.$$

$$g_*^{n+1} \xrightarrow{\text{weighted SVD}} T_r(g_*^{n+1})$$

$$f^{n+1} = f_*^{n+1} + T_r(g_*^{n+1})$$

3. semi-Lagrangian (combine x & t)

Back to Sec 4.

Where did particle (x, v) come from?

$$\frac{dx}{dt} = v \quad \frac{dv}{dt} = -E(t, x) \Rightarrow \frac{d}{dt} f(t, x(t), v(t)) = 0.$$

$$\text{phase trans: } \partial_t f + v \partial_x f = 0.$$

$$(f \text{ fix } v) \Delta x = v \Delta t.$$

$$\text{Then } \tilde{f}(t + \Delta t, x, v) = f(t, x - \Delta t v, v)$$

$$v \text{ trans: } \partial_t f - E(x) \partial_v f = 0.$$

$$f(t + \Delta t, x, v) = \tilde{f}(t + \Delta t, x, v + \Delta t E(t, x))$$

$x_k - \Delta t v_k$ not on mesh $\Rightarrow$ interpolation (linear)

$$\tilde{f}(x_k, v_k) \approx f(x_k, v_k) (1 - |\Delta t v_k|) + f(x_{k-1}, v_k) \max(0, \Delta t v_k) + f(x_{k+1}, v_k) \max(0, -\Delta t v_k)$$

rank $\leq 3r \Rightarrow$ SVD Truncate.

Then in sec 6

$$f^n \rightarrow f_*^{n+1} \quad p, j, k \dots ?$$

semi-L $\Rightarrow p_k, j_k$

$$\text{Conservation} \Rightarrow p_f, j_f \quad \Rightarrow \text{modify } f_* \quad (u_k = \frac{j_k}{p_k} \Rightarrow u_k \Rightarrow j_k)$$

Partial Conservation

Fix for DLR.

$$\left\{ \begin{array}{l} \text{Modify projection conditions. (A)} \\ \text{Enlarge Conservation (B)} \\ \text{Macro-Micro (C)} \end{array} \right.$$

A: Modify projection conditions.

Standard DLR $\partial_t f = \text{RHS}(f)$

$$f_r(t) \in M_r \quad \partial_t f_r = \arg \min_{g \in T_{f_r, M_r}} \|g - \text{RHS}(f_r)\|_{L^2}$$

$$\partial_t f_r(t) \in T_{f_r, M_r} \quad \partial_t f_r = \text{RHS}(f_r)$$

$$\Rightarrow \partial_t f_r = P_{T_{f_r, M_r}} \text{RHS}(f_r) \Leftrightarrow \langle R, Sf \rangle_{L^2} = 0 \text{ for } \forall Sf \in T_{f_r, M_r}$$

Why not conserve?

$$\langle R, 1 \rangle_v = 0 \quad \langle R, v \rangle_v = 0 \quad \langle R, |v|^2 \rangle_v = 0 \quad \dots \times$$

$$\Rightarrow A: \quad \text{Add } \langle R, 1 \rangle_v = 0 \quad \langle R, v \rangle_v = 0 \quad \langle R, |v|^2 \rangle_v = 0.$$

$$\int 1 = \infty \Rightarrow \text{weighted } L^2.$$

How to add?

$$\text{eg. } e^{-\frac{v^2}{2}}$$

$$f(t, x, v) = f_{0v}(v) \sum_{i,j=1}^k X_i(t, x) S_{ij}(t) V_j(t, v)$$

$$\text{Fix } u_1(v) \propto 1 \quad u_2(v) \propto v \quad u_3(v) \propto v^2 - 1$$

$$\text{other basis } w_4(t, v) \dots w_r(t, v)$$

$$\text{test } \left(\frac{v}{f_{0v}}, \partial_t f - D[f] \right)_{xv} = 0 \quad \forall v \in T_{f, M}$$

B: Modify basis-space.

$$\text{Augmented BUA} \Rightarrow \hat{V} = \text{orth} [V^n, V^{\text{pred}}, 1, v]$$

$$\partial_t f = F[f] \quad f^n(x, v) = \sum x_i^n S_{ij}^n v_j^n$$

$$\Psi(v) = \begin{pmatrix} 1 \\ v \end{pmatrix}$$

$$\Phi^n(x) = \langle f^n(x, \cdot), \Psi \rangle_v = \begin{pmatrix} p^n(x) \\ j^n(x) \end{pmatrix}$$

$$\text{Keep } \Phi^{n+1}(x) = \Phi^n(x) + \Delta t \langle F[f^n], \Psi \rangle_v$$

Pre-processing : $1. v \in \text{Span}\{v_1^0, \dots, v_{r_0}^0\}$
 $f = f_{0v}(v) X S V^T$

Complete conservative augmented BUG

Step 1: $K^n = X^n S^n$.

(Same as augBUG) $K^{n+1} = K^n + \Delta t \langle F(f^n), V^n \rangle_v$
 $\hat{X} = \text{orth}[X^n, K^{n+1}]$

Step 2: $L^n \dots$

$\hat{V} = \text{orth}[V^n, L^{n+1}]$

$1, v \in \text{Span}(V^n) \subset \text{Span}(\hat{V})$

Step 3: $M = \langle \hat{X}, X^n \rangle_x \quad N = \langle \hat{V}, V^n \rangle_v$

(Same) $\hat{S}^n = M S^n N^T$

$\hat{X} \hat{S}^n \hat{V}^T = X^n S^n (V^n)^T = f^n$

Step 4: $\hat{S}^{n+1} = \hat{S}^n + \Delta t \langle F(f^n), \hat{X} \hat{V}^T \rangle_{xv}$

$\Rightarrow f^{n+1} = \hat{X} \hat{S}^{n+1} \hat{V}^T$

Step 5. (conventional truncate)

$V_{\text{cons}} = \text{span}\{1, v\} \quad P_{\text{cons}}$

$\hat{f}^{n+1} = f_{\text{cons}}^{n+1} + f_1^{n+1} \quad (\langle f_1^{n+1}, 1 \rangle = \langle f_1^{n+1}, v \rangle = 0)$

SVD Truncate for f_1^{n+1}

C. $\partial_t f + v \partial_x f + E \partial_v f = Q(f) \quad (*)$

$f = \underbrace{N}_{\text{macro conserve}} + \underbrace{g}_{\text{micro normal DLR}}$

$\int N dv = \int f dv$

$\int v N dv = \int v f dv$

$\int \frac{v^2}{2} N dv = \int \frac{v^2}{2} f dv$

Set $N(x, v, t) = a_0(x, t) q_0(v) + a_1(x, t) q_1(v) + a_2(x, t) q_2(v)$

i.e. $\mathcal{Q} = \text{Span}\{q_0, q_1, q_2\}$.

if $(q_0, q_1, q_2) = (1, v, v^2)$. not approved cuz $\uparrow$ when $v \uparrow$.

$\Rightarrow \{w, vw, v^2 w/2\}$

choose $p_0(v), p_1(v), p_2(v) \Rightarrow \int w p_i p_j dv = \delta_{ij}$

$q_i(v) = w(v) p_i(v)$ the $\langle q_i, q_j \rangle_{w^{-1}} = \delta_{ij}$

then $N(x, v, t) = \sum_{i=0}^2 q_i f_i = w \sum p_i f_i$
 $\langle \sum p_i, w p_k \rangle_{w^{-1}} = \langle f, p_k \rangle$.

$\mathcal{Q} = \text{Span}\{w p_0, w p_1, w p_2\}$.

$N = P_{\mathcal{Q}} f \quad g = f - N$.

How to update $U = (f_0, f_1, f_2)^T \quad g_r = X S V^T \quad ?$

$f_r = N + g_r$.

$\partial_t f = D I E, f \quad f - v \partial_x f - E \partial_v f + Q(f)$

$\partial_t (N + g) = D I E, N + g$

$\Rightarrow \partial_t N = P_{\mathcal{Q}} D I E, N + g$

$N = w P^T U$

p_0, p_1, p_2 on (x) : $\partial_t U + v \partial_x U + a_2 \partial_x f_3 e_2 + E D_v U = 0 \quad (M)$

$$P_{\frac{1}{2}} \text{ on } (x): \partial_t q = P_{\frac{1}{2}}^\perp DLE, N+g \quad (m)$$

$$g \text{ offer } f_3 = \langle P_3, f \rangle$$

$$f = N+g \Rightarrow \langle P_3, N \rangle = 0.$$

$$\Rightarrow f_3 = \langle P_3, f \rangle = \langle P_3, g \rangle$$

$$\text{i.e. } f_3^n = \sum_{i,j} X_i^n S_{ij}^n \int P_3 V_j^n (v) dv.$$

N offer .

$$P_{\frac{1}{2}}^\perp DLE, N, g \text{ include } v \partial_x N, E \partial_v N, Q(N+g)$$

$$\text{Functions: } p = \int f dv \quad J = \int v f dv \quad K = \int \frac{v^2}{2} f dv$$

$$\text{satisfying } \partial_t p + \partial_x J = 0$$

$$\partial_t J + \partial_x (2K) = pE$$

$$\partial_t K + \partial_x \left(\int \frac{v^3}{2} f dv \right) = JE$$

↳ offered by g

A whole temporal step

$$\text{Macro: } U^n \cdot g_r^n = X^n S^n (V^n)^T \cdot E^n$$

$$N^n = w [p_0 f_0^n + p_1 f_1^n + p_2 f_2^n]$$

$$f_3^n = \sum_{i,j} X_i^n S_{ij}^n \langle P_3, V_j^n \rangle v$$

$$\text{update } U^n: \frac{f_0^{n+1} - f_0^n}{\Delta t} = \partial_x (-b_0 f_0^n - a_0 f_0^n)$$

$$\frac{f_1^{n+1} - f_1^n}{\Delta t} = -a_1 \partial_x f_1^n - b_1 \partial_x f_1^n - a_0 \partial_x f_0^n + E^* d_{10} f_0^n$$

$$\frac{f_2^{n+1} - f_2^n}{\Delta t} = -a_2 \partial_x f_2^n - b_2 \partial_x f_2^n - a_1 \partial_x f_1^n + E^* (d_{20} f_0^n + d_{21} f_1^n)$$

$$U^{n+1} \Rightarrow N^{n+1} \dots$$

$$\text{Micro: } R_g^n = P_{\frac{1}{2}}^\perp DLE^*, N^n, g_r^n$$

$$g_r^n = X^n S^n (V^n)^T$$

$$\partial_t q = P_{\frac{1}{2}}^\perp DLE, N, g$$

use standard DLR (projector-splitting) on it.

$$[\text{Modify: QR for } L^n [w p_0, w p_1, w p_2, L^{n+1}] \xrightarrow{\text{weighted QR}} [w p_0, w p_1, w p_2, V^{n+1} R]]$$

↳ Guarantee that $V \perp \mathbb{F}$

$$g_r^{n+1} = X^{n+1} S^{n+1} (V^{n+1})^T$$

$$f_r^{n+1} = N^{n+1} + g_r^{n+1}$$

6.2 Asymptotic Preserving (AP) (渐进保持)

Kinetic with strong-collision, $\epsilon \rightarrow 0$ if low rank $\checkmark$?

$$\text{i.e. } f(t, x, v) = p(t, x) + \epsilon g(t, x, v)$$

(fix $\Delta t, \Delta x$) $\epsilon \rightarrow 0$, p diffusion equation

1° diffusion limit.

$$\partial_t f + \frac{1}{\epsilon} v \cdot \nabla_x f = \underbrace{\frac{\sigma_s(x)}{\epsilon^2} (p - f)}_{\text{isotropic scattering}} + \underbrace{G(x)}_{\text{absorption}} \quad p(t, x) = \langle f \rangle_v = \frac{1}{4\pi} \int_{S^2} f(v) dv$$

Transport under diffusive scaling $\frac{\text{mean free path}}{\text{microscopic length}}$

$$\text{collision } Q(f) = \sigma_s (p - f) \quad p(x, t) = \langle f(x, \cdot, t) \rangle_v$$

Supposing that $f(v_1) = 8 \quad f(v_2) = 4 \quad f(v_3) = 2$ on different directions

the $f \rightarrow p$ (average) if $\epsilon \rightarrow 0$

$$\text{i.e. } f(x, v, t) \approx p(x, t) \quad f \rightarrow p$$

$$\text{i.e. } f = p + \epsilon g$$

$$\Rightarrow \partial_t(P + \epsilon g) + \frac{1}{\epsilon} V \cdot \nabla_x (P + \epsilon g) = -\frac{\sigma_s}{\epsilon} g - \sigma_a (P + \epsilon g) + G \quad (*)$$

Macro equation: average for V . ($\langle g \rangle_V = 0$, $\langle V \rangle_V = 0$)

$$\partial_t P + \nabla_x \cdot \langle V g \rangle_V = -\sigma_a P + G$$

Micro equation: push $1-P$ on $(*)$ and divided by ϵ

$$\partial_t g + \frac{1}{\epsilon} (1-P) (V \cdot \nabla_x g) + \frac{1}{\epsilon^2} V \cdot \nabla_x P = -\frac{\sigma_s}{\epsilon^2} g - \sigma_a g$$

$$\times \epsilon^2: \quad \epsilon^2 \partial_t g + \epsilon (1-P) (V \cdot \nabla_x g) + V \cdot \nabla_x P = -\sigma_s g - \epsilon^2 \sigma_a g$$

$$\epsilon \rightarrow 0: \quad g = -\frac{1}{\sigma_s} V \cdot \nabla_x P \quad \text{扩散闭合关系 (diffusive closure)}$$

$$\text{bring back to macro: } \langle V g \rangle_V = -\frac{1}{\sigma_s} \langle V \otimes V \rangle_V \cdot \nabla_x P = -\frac{1}{3\sigma_s} \nabla_x P$$

$$\text{Thus } \partial_t P - \nabla_x \cdot \left(\frac{1}{3\sigma_s} \nabla_x P \right) = -\sigma_a P + G$$

Why low-rank?

$$f = P + \epsilon g$$

$$g \approx -\frac{1}{\sigma_s} \sum_{k=1}^d \partial_{x_k} P(x) V_k$$

$$\hookrightarrow g(x, V) \approx \sum_{k=1}^d X_k(x) V_k(V)$$

Methods under diffusion limit

A: Direct DLR

$$f_r = X S V^T$$

$$P_{T_r} M_r = P_x + P_v - \underbrace{P_x P_v}_{\hookrightarrow \text{Inverse step}}$$

$\Rightarrow$ proved by [44]: direct DLR X

B: DLR only for micro:

$$\text{Keep } p^n(x) \text{ and } g_r^n(x, V) = \sum_{i,j=1}^f X_i^n S_{ij}^n V_j^n(V)$$

$$f_r^n = p^n + \epsilon g_r^n$$

$$\text{Macro: } \langle V g_r^n \rangle_V = \sum_{i,j} X_i^n S_{ij}^n \langle V V_j^n \rangle_V$$

$$\frac{p^{n+1} - p^n}{\Delta t} + \nabla_x \cdot \langle V g_r^n \rangle_V = -\sigma_a p^n + G^*$$

$$\text{Micro: project splitting for } \partial_t g + \frac{1}{\epsilon} (1-P) (V \cdot \nabla_x g) + \frac{1}{\epsilon^2} V \cdot \nabla_x P = -\frac{\sigma_s}{\epsilon^2} g - \sigma_a g$$

$$\text{i.e. } \partial_t g = -\frac{1}{\epsilon} (1-P) (V \cdot \nabla_x g) - \frac{1}{\epsilon^2} (V \cdot \nabla_x P + \sigma_s g) - \sigma_a g$$

$$\epsilon \rightarrow 0, \quad V \cdot \nabla_x P + \sigma_s g \approx 0.$$

if direct discretize:

$$(\text{using } \frac{1}{\epsilon^2} \text{ only}) \quad g^{n+1} = g^n - \frac{\Delta t}{\epsilon^2} (V \cdot \nabla_x P^n + \sigma_s g^n)$$

$\nearrow$ exploded if fix Δt .

$$\text{IMEX: } O(\frac{1}{\epsilon}) \quad g^n, \quad O(\frac{1}{\epsilon^2}) \quad g^{n+1}, \quad \text{Absorption } g^n, \quad \text{Macro } P^*(g^n)$$

$$\frac{g^{n+1} - g^n}{\Delta t} + \frac{1}{\epsilon} (1-P) (V \cdot \nabla_x g^n) + \frac{1}{\epsilon^2} V \cdot \nabla_x P^n + \frac{1}{\epsilon^2} V \cdot \nabla_x P^* = -\frac{\sigma_s}{\epsilon^2} g^{n+1} - \sigma_a g^n.$$

$$\text{then } \hat{g}^{n+1} = \frac{\dots}{\epsilon^2 + \Delta t \sigma_s}$$

$$\epsilon \rightarrow 0 \text{ to get } g^{n+1} \rightarrow -\frac{1}{\sigma_s} V \cdot \nabla_x P^* \text{ bring back to } P:$$

$$\frac{p^{n+1} - p^n}{\Delta t} - \nabla_x \cdot \left(\frac{1}{3\sigma_s} \nabla_x P^* \right) = -\sigma_a P^* + G^*$$

C: projector-splitting $\Rightarrow$ BUG

$$g \rightarrow -\sigma_s^{-1} \Delta \times P$$

$$\hookrightarrow \frac{p^{n+1} - p^n}{\Delta t} - \frac{1}{2} \Delta \times (\sigma_s^{-1} \Delta \times P^n) = 0. \quad (\text{扩散扩散方程})$$

(satisfying CFL condition $\Delta t \leq C(\epsilon \Delta x + \sigma_0 \Delta x^2)$)

D: SAT $g^{n+1,*} = \text{IMEX}$

$$g^{n+1} = T_\eta(g^{n+1,*})$$

$$\| -\frac{1}{3\sigma_s} \nabla_x P - \langle \Omega T_\eta(g) \rangle_\Omega \| \leq \epsilon + C\delta \text{ truncate}$$

E: (not macro - micro) GAP

$$f_r = XSV^T$$

P, g

$$\text{fix } X^n, \quad l(t) = X^n L(t)^T \quad (= f_r^n)$$

$$\partial_t l = P_X^n L(t), \quad l(t_n) = f_r^n$$

$$\Rightarrow V^{n+1}$$

$$\text{fix } V^{n+1} \quad \partial_t k = P_{V^{n+1}} L(k) \quad k(t_n) = P_{V^{n+1}} f_r^n$$

$$\Rightarrow X^{n+1}, S^{n+1}$$

2° Fluid limit

$$\text{BGK equation: } \partial_t f + v \cdot \nabla_x f = \frac{V}{\epsilon} (M[f] - f)$$

$$\text{partial Maxwellian: } M[f](x, v) = \frac{P(x)}{(2\pi)^{d/2}} \exp\left(-\frac{|v - u(x)|^2}{2}\right) \quad \text{难以低维表示.}$$

$$f = M[f] + O(\epsilon)$$

$$U = (P, \rho u, E)^T \rightarrow \text{Euler equation (守恒分布)} \Rightarrow f = M[U]g.$$

$$\text{eg. } M(x, v) = \frac{P(x)}{(2\pi)^{d/2}} \exp\left(-\frac{|v - u(x)|^2}{2}\right)$$

$$\text{including } e^{v \cdot u(x)} \text{ can't be written as } \sum_{i=1}^N X_i(x) V_i(v) \quad (r.v.)$$

A Weakly compressible Maxwellian (守恒, 弱可压缩 BGK)

$$(\text{When } u \ll 340 \text{ m/s.}) \quad e^{v \cdot u - |u|^2/2} \approx 1 + v \cdot u + \frac{1}{2}(v \cdot u)^2 - \frac{1}{2}|u|^2$$

$$\Rightarrow M(x, v) \approx \frac{P(x)}{(2\pi)^{d/2}} e^{-|v|^2/2} [1 + v \cdot u(x) + \dots]$$

$$\text{numerical: } f_r(x, v, t) = \sum_{i,j=1}^K X_i(x, t) S_{ij}(t) V_j(v, t).$$

$$\text{DLR (RHS} = \frac{V}{\epsilon} (M^n - f_r^n))$$

B. 完全可压缩 BGK 乘法分解

u not small?

$$M[f](x, v) = \frac{P(x)}{(2\pi)^{d/2}} \exp\left(-\frac{|v - u(x)|^2}{2}\right)$$

$$f = M g$$

$$\text{DLR for } g \Rightarrow f_r = M[g, u] g_r$$

$$\text{BGK: } \partial_t f + v \cdot \nabla_x f = \frac{V}{\epsilon} (M - f)$$

$$f = M g$$

$$\partial_t f = M \partial_t g + g \partial_t M$$

$$(\text{代入 BGK}) \Rightarrow \partial_t g = -v \cdot \nabla_x g - (\partial_t \log M + v \cdot \nabla_x \log M) g + \frac{V}{\epsilon} (1 - g) = -v \cdot \nabla_x g - \Gamma[U]g + \frac{V}{\epsilon} (1 - g) \quad \checkmark$$

$$(\text{update } g) \Rightarrow \text{DLR/projector-splitting}$$

$$(\text{update } U) \quad U = (P, \rho u, E)^T$$

$$\Rightarrow \text{keep } \Gamma: \quad U = (P, \rho u)^T \quad (\text{乘积积分}) \Rightarrow \partial_t U + \nabla_x \cdot \langle v \phi(v) M[U] g_r \rangle_v = 0 \quad \checkmark$$

$$\text{A time step: } U^n, \quad g_r^n = X^n S^n (V^n)^T$$

$$M^n = M[P^n, u^n, T^n]$$

$$F_v^n = \langle v \phi(v) M^n g_r^n \rangle_v$$

$$U^{n+1} = U^n - \Delta t \cdot \nabla_x \cdot F_v^n$$

$$\Rightarrow \partial_t \log M \cdot \nabla_x \log M$$

$$\partial_t g = -v \cdot \nabla_x g - (\partial_t \log M + v \cdot \nabla_x \log M) g + \frac{V}{\epsilon} (1 - g)$$

$$\Rightarrow \text{projector-splitting}$$

$$(\text{ } g^{n+1} \text{ for } \frac{V}{\epsilon} (1 - g))$$

$$f_r^{n+1} = M[U^{n+1}] g_r^{n+1}$$

Prop: g - low-rank

Proof: Chapman-Enskog 展开: $f = M + \epsilon f_1 + O(\epsilon^2)$

$$(\text{修正: } f_1 = M g_1(x, v))$$

$$g_1 \sim \text{poly}(v \cdot u)$$

$$\Rightarrow f = M(1 + \epsilon g_1 + O(\epsilon^2))$$

$$g = \frac{f}{M} = 1 + \epsilon g_1 + O(\epsilon^2) \rightarrow 1$$

$$\partial_t f = v \cdot \nabla_x f = \frac{V}{\epsilon} (M - f)$$

$$\phi(v) = (1 - v^{|u|/2})^T$$

$$U = \langle \phi f \rangle_v = \begin{pmatrix} P \\ E \end{pmatrix}$$

$$\partial_t \langle \phi f \rangle_v + \nabla_x \cdot \langle v \phi f \rangle_v = \frac{V}{\epsilon} \langle \phi (M - f) \rangle_v$$

$$\langle \phi M \rangle_v = \langle \phi f \rangle_v = U$$

$$\partial_t U + \nabla_x \cdot \langle v \phi f \rangle_v = 0.$$

$$f = M g$$

$$\Rightarrow \partial_t U + \nabla_x \cdot F_v = 0, \quad F_v := \langle v \phi M g \rangle_v.$$

Problems in B: Integral $\langle v V_j M \rangle_v$ $\langle v \otimes v V_j M \rangle_v$.

$\Rightarrow$ Solution 152] convolution

$$\begin{aligned} \langle v V_j M \rangle_v &= \frac{P(x)}{(2\pi)^{d/2}} \int v V_j(v) e^{-|v-u(x)|^2/2} dv \\ \langle v \otimes v V_j M \rangle_v &= \frac{P(x)}{(2\pi)^{d/2}} \int v \otimes v V_j(v) e^{-|v-u(x)|^2/2} dv \\ \Rightarrow 152]: \quad C_j^{(1)}(u) &= \int v V_j(v) e^{-|v-u|/2} dv \quad v_j \\ C_j^{(2)}(u) &= \int v \otimes v V_j(v) e^{-|v-u|/2} dv \quad v_j \\ C_j^{(1)}(u(x_i)) & \quad C_j^{(2)}(u(x_i)) \\ \Rightarrow \langle v M q_r \rangle_v &= \frac{P}{(2\pi)^{d/2}} \sum_{i,j} \chi_i S_{ij} C_j^{(1)}(u(x_i)) \\ & \quad O(\gamma n^3) \end{aligned} \quad u=u(x) \text{ differentiate in each } x.$$

Space Discretize

1° Low-rank K-step

$$\partial_t K + C_x \partial_x K + C_y \partial_y K + B(x) K = \frac{P}{\epsilon} (\bar{V} - K)$$

$$(C_x)_{jl} = \langle v_x V_j V_l \rangle_v \quad (C_y)_{jl} = \langle v_y V_j V_l \rangle_v \quad \text{对称, 可对称化.}$$

$$\Rightarrow K^{n+1} = \frac{K^n - \Delta t D(K^n) + \frac{\Delta t P^n}{\epsilon} \bar{V}}{1 + \frac{\Delta t P^n}{\epsilon}}$$

2° Macro moment equation

$$U = (P, P u)^T$$

$$\partial_t U + \nabla_x \cdot F(U, g_r) = 0$$

$$F(U, g_r) = \langle v \phi(u) M^T U \rangle g_r$$

A time step

$$U^n = (P^n, P^n u^n)^T \quad g_r^n = X^n S^n (V^n)^T$$

1° $M^n = M^T P^n, u^n$

$$\begin{aligned} \langle v V_j^n M^n \rangle_v & \quad \langle v \otimes v V_j^n M^n \rangle_v \\ \Rightarrow F(U^n, g_r^n) \end{aligned}$$

2° Using Nessyahu - Tadmor $\rightarrow \partial_t U + \nabla_x \cdot F(U, g_r) = 0$

$$U^{n+1} = (P^{n+1}, P^{n+1} u^{n+1})^T$$

3° $\Gamma = \frac{1}{M} (\partial_t M + v \cdot \nabla_x M)$

$$(\frac{1}{M}) \quad \Gamma = \Gamma_1(x) + v \cdot \Gamma_2(x) + (v \otimes v) \cdot \Gamma_3(x)$$

$$4° \quad \partial_t g = -v \cdot \nabla_x g - \Gamma g + \frac{v}{\epsilon} (1-g)$$

$$\Rightarrow \text{IMEX K-S-L} \quad \text{--- BAK.}$$

$$g_r^{n+1} = X^{n+1} S^{n+1} (V^{n+1})^T$$

$$5° \quad f_r^{n+1} = M^T U^{n+1} g_r^{n+1} \\ (U^{n+1}, X^{n+1}, S^{n+1}, V^{n+1})$$

C. Vlasov - Ampère - Fokker - Planck 中微子分解.

$$\partial_t f + v \cdot \nabla_v f + \frac{1}{\epsilon} E \cdot \nabla_v f = \frac{1}{\epsilon} \nabla_v \cdot (v f + v v f)$$

$$E \text{ Updated by Ampère } \partial_t E = -J \quad J = \langle v f \rangle_v$$

$$\text{Rewrite: } \partial_t f + v \cdot \nabla_v f = \frac{1}{\epsilon} \nabla_v \cdot [M_E \nabla_v (M_E^{-1} f)] \quad \text{where } M_E = \frac{1}{(2\pi)^{d/2}} \exp(-\frac{|v-E(x)|^2}{2})$$

$$\text{Hard Collision Limit: } f \rightarrow P(x) M_E(x, v)$$

$$\Rightarrow f = M_E g \quad g_r = X S V^T \quad (g \rightarrow P(x) \text{ in limit})$$

$$g \text{ update: } \partial_t g + v \cdot \nabla_x g = -\Gamma_E g + \frac{1}{\epsilon} M_E^{-1} \nabla_v \cdot (M_E \nabla_v g) \Rightarrow \text{K-S-L update } (\Gamma_E = M_E^{-1} (\partial_t M_E + v \cdot \nabla_x M_E))$$

LC to prove that $f = M g$ is appropriate not only for BAK.

D. 插值型 DLR (only calculates on a few points)

$$\partial_t f + v \cdot \nabla_x f = \frac{1}{\epsilon} (M[f] - f)$$

$$M[f](x, v) = \frac{p(x)}{(2\pi T(x))^{3/2}} \exp\left(-\frac{|v - u(x)|^2}{2T(x)}\right)$$

$$\text{Set } f_r = XSV^T$$

$$\text{RHS: } h(f_r) = -v \cdot \nabla_x f_r + \frac{1}{\epsilon} (M[f_r] - f_r)$$

$$\text{Normal DLR: compute } \langle M[f_r] V_j \rangle_v = \int M[f_r](x, v) V_j(v) dv \quad X.$$

$$\text{Fix: DEIM choose } X = (x_{d_1}, \dots, x_{d_r}), \quad V = (v_{\beta_1}, \dots, v_{\beta_r})$$

$\Rightarrow$ Oblique projection

6.3. Positivity

How Low-rank keep $f(x, v, t) \geq 0$.

1° why $f < 0$?

2° Introduce g

$$f = g^2 \quad (g^*g = |g|^2)$$

$$\Rightarrow g_r = XSV^T$$

$$f_r = g_r^2 \Rightarrow f_r \geq 0.$$

3° e.g. Vlasov Equation $\partial_t f + v \cdot \nabla_x f + \frac{1}{m} F \cdot \nabla_v f = 0$.

$$\text{逆推算子 } D = \partial_t + v \cdot \nabla_x + \frac{1}{m} F \cdot \nabla_v$$

$$\Rightarrow Df = 0. \quad \text{i.e. } 2g Dg = 0.$$

$$\Rightarrow \partial_t g + v \cdot \nabla_x g + \frac{1}{m} F \cdot \nabla_v g = 0.$$

4° For Vlasov-Maxwell

$$\text{for } S: \partial_t g_s + v_s \cdot \nabla_x g_s + \frac{q_s}{m_s} (E + v_s \times B) \cdot \nabla_{v_s} g_s = 0.$$

$$g_{s,r} \quad f_{s,r} = g_{s,r}^2$$

5° Limitation

$$(1) \text{ e.g. } \partial_t f = \Delta f$$

$$f = g^2 \Rightarrow 2g \partial_t g = 2g \Delta g + 2|\nabla g|^2$$

$$\partial_t g = \Delta g + \underbrace{\frac{|\nabla g|^2}{g}}_X$$

(2) low-rank problem.

$$g_r^2 = \sum_{i,j=1}^r A_i A_j B_i B_j$$

Sec 7 High-efficiency computing. (CPU & GPU)

Sec 8 Summary and open problems

1° math basis:

$$\text{error estimate } \|F(f) - F(g)\| \leq L \|f - g\|$$

$$L \sim \frac{1}{\Delta x} ? \quad L \sim \frac{1}{\epsilon}$$

Suppose that RHS mainly in a low-rank space i.e. $\|(I - P_{T_f, M_r}) F(f_r)\| \leq \delta_r$

2° Expressing ability (choose low-rank structure)

3° Solving Efficiency

4° Physical Structure and Application

4.4 Stability.

Linear Stability: $\partial_t u + A \partial_x u = 0$

$$\text{discretize: } u^{n+1} = u^n + \Delta t \text{ RHS}_\Delta(u^n)$$

$$\Phi_{\Delta t}(u) = u + \Delta t \text{ RHS}_\Delta(u)$$

$$L^2 \text{ stable: } \|\Phi_{\Delta t}(u)\|_2 \leq \|u\|_2$$

SA7 V:

$$F_r^n \rightarrow \tilde{F}_r^{n+1} = \Phi_{\Delta t}(F_r^n) \xrightarrow{\text{SVD}} F_r^{n+1} = T_r(\tilde{F}_r^{n+1})$$

$$\|T_r(\tilde{F})\|_F \leq \|\tilde{F}\|_F^2 \leq \|F_r^n\|_F$$

DLR: $F_r = USV^T$

Projecting space varies with solution

K-step fix V

S-step fix U, V

L-step fix U

Each step is linear to current unknown variables.

↳ QR only transfers. SVD ↓

Problem of projector - splitting

$$P_T Z = P_U Z + Z P_V - P_U Z P_V$$

K-step: $Z P_V$

S-step: $-P_U Z P_V$

L-step: $P_U Z$

$$R_\Delta(F) = \text{Transport}(F) + D_\Delta(F)$$

$$\langle F, D_\Delta(F) \rangle_\Delta \leq 0.$$

↳ $-P_U D_\Delta(F) P_V$ might cause numerical diffusion → anti-diffusion

A

B

Discretize then DLR vs DLR then Discretize

Plan A: $\dot{F}_h = R_\Delta(F_h)$

$$\dot{F}_r = P_{T_r, M_r} R_\Delta(F_r)$$

e.g. $\partial_t f + v \partial_x f = 0$

$$F_{ij}(t) = f(t, x_i, x_j)$$

Upwind: $\dot{F} = -D_x F \text{ diag}(v) := R_\Delta(F)$

$$F_r = USV^T \rightarrow \dot{F}_r = P_{T_r, M_r}^\Delta R_\Delta(F)$$

Plan B: $f_r(t, x, v) = XSV^T$

$$R(f_r) = -v \partial_x f_r$$

$$\partial_t f_r = P_{T_r, M_r} R(f_r)$$

$$P_{T_r, M_r} R_\Delta(F_r) \neq [P_{T_r, M_r} R(f_r)]_\Delta$$